1. Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges or diverges using the integral test.

2. The exact sum of some series can be found. For instance, $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$. Use this fact to find the exact sum of the following series. [Give decimal approximations as well.]

   (a) $\sum_{n=2}^{\infty} \frac{1}{n^4}$ (b) $\sum_{n=2}^{\infty} \frac{1}{(n+1)^4}$ (c) $\sum_{n=1}^{\infty} \frac{1}{(2n)^4}$

3. (calculator) Find the partial sum s_4 of the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$. Estimate the error in using s_4 as an approximation of the series.

4. Determine whether the following series converge or diverge using direct comparison. To receive full credit you must (1) explain what you are comparing the terms of the given series to, (2) explain what the series you are comparing to converges/ diverges, (3) make a conclusion.

a) $\sum_{n=1}^{\infty} \frac{1+\cos n}{e^n}$

b) $\sum_{n=1}^{\infty} \frac{n+1}{n\sqrt{n}}$

5. Test the following series for convergence or divergence.

a) $\sum_{n=1}^{\infty} (-1)^n \frac{3n-1}{2n+1}$

b) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^3}{n^4+1}$